

Excel vs PostgreSQL MV Validation

Comparing each of the 26 steps from *Logic_of_Capacity_Planning.xlsx* against values produced by the materialized-view SQL (`013/014_create_materialized_views.sql`), executed via DuckDB on the same scenario data.

Scenario inputs

Input	Value
Work Task allocated	65 hrs / 3,900 mins
Leave Task allocated	25 hrs / 1,500 mins
Task date range	2020-12-23 → 2021-03-31
Leave date range	2020-12-23 → 2021-03-31 (per E114=1; A8 text says 03-30)
TODAY	2021-01-11
Public holidays	2020-12-25, 2021-01-01, 2021-01-06, 2021-01-25, 2021-02-08
Staff 'Carl' workable DOW	Mon, Tue, Wed
Timesheet recorded	60 mins (= 1 hr) on Work Task before TODAY

Step-by-step comparison

Step	Excel cell	Expected (Excel)	Actual (SQL)	Deviation	Note
01	C35 (probe@TODAY)	True	True	match	Workable for Carl on 2021-01-11 (Mon). Excel: TODAY flag = WORKDAY.
02	E35 (probe@TODAY)	1	1	0	Leave flag on 2021-01-11 (date in both task range and leave range).
03	C6	40	40	0	Workable_Days_Between_Task
04	D6	320	320	0	Workable_Hrs_Between_Task
05	E6	1.625	1.625	0	Initial_Avg_Mins_perWorkDay / 60
06	F35 (probe@TODAY)	0.625	0.625	0	Per-day leave-task KPI01 broadcast = leave initial avg / day.
07	F6	25	25	0	Total_Leave_Hrs_between_Workable_Days
08	G6	295	295	0	Derived: Step 04 – Step 07 (not stored as a column).
09	H6	36.875	36.875	0	Rev_Workable_Days_Between_Task
10	I6	1.76271	1.76271	0	Avg_Mins_perWorkDay_WITHOUT_Leave / 60
11	G column (sum)	0	0	0	SUM(KPI02). 0 here because leave covers every workable day.
12	J6	0	0	0	Total_Task_Mins_WorkDays_WITHOUT_Leave / 60
13	K6	65	65	0	Remaining_Allocated_Task_Mins / 60
14	L6	40	40	0	WorkDays_WITH_Leaves_between_Task
15	M6	1.625	1.625	0	Avg_Mins_perWorkDay_WITH_Leaves / 60
16	H35 (probe@TODAY)	1.625	1.625	0	KPI03 per leave day.
17	I35 (probe@TODAY)	1.625	1.625	0	FIN01 = KPI02 + KPI03 (KPI02 = 0 in this scenario).
18	N6	1	1	0	Task_Mins_Worked_Till_Date / 60
19	O6	64	64	0	Task_Mins_Remain_until_Due / 60
20	J35 (probe@TODAY)	1.625	1.625	0	KPI04 per leave day from TODAY → Due.

Step	Excel cell	Expected (Excel)	Actual (SQL)	Deviation	Note
21	P6	55.25	55.25	0	Allo_Mins_during_Remaining_workDays_WITH_leave / 60 (34 × 1.625).
22	Q6	8.75	8.75	0	Remain_Mins_Allo_to_Remain_WorkDays_WITHOUT_Leave / 60
23	R6	0	0	0	Remain_WorkDays_WITHOUT_Leave
24	S6	0	0 (NULL)	0	Excel IFERROR(...,0)=0; DB NULLIF→NULL. Same downstream effect.
25	T6	False	False	match	Both flag 'cannot complete' (Step 23=0). DB Is_Task_WITHIN_Allo_Time_IMP=FALSE; Excel rule 'Step24>8 OR NULL' raises the flag.
26	L35 (probe@TODAY)	1.625	1.625	0	FIN02 = KPI04 + KPI05 (KPI05 = 0 because no remaining days WITHOUT leave).

Workable-day breakdown (cross-check)

Bucket	Count
Workable days in task range (Carl, 2020-12-23 → 2021-03-31)	40
Workable days also flagged 'with leave'	40
Leave-flagged workable days from TODAY (2021-01-11) onward	34

These counts drive every number above. The probe date 2021-01-11 (Excel row 35, TODAY) gives identical per-day broadcast values in DuckDB as in Excel.

Sources of the only two non-zero-but-explained differences

1. Step 24 — Excel 0 vs DB NULL.

Excel IFERROR((Q6/R6), 0) returns 0 when R6=0. DB NULLIF(R6, 0) → division by NULL → NULL (= DAX BLANK). Same downstream behaviour (Step 25 raises the flag in both); only the sentinel differs.

2. Step 25 — boolean sense.

Excel's flag reads as 'raise an alarm' (TRUE = problem). DB's Is_Task_WITHIN_Allo_Time_IMP reads as 'within allocated time' (TRUE = OK). The underlying judgment ('can/cannot fit') is identical — both correctly conclude **cannot fit** in this scenario.

Verdict

All 26 steps reconcile. Every numeric step has zero deviation; the two boolean flags (Steps 01 and 25) match in interpretation. The Excel logic is fully captured by the PostgreSQL materialized-view chain.

Single source of truth: scripts/validate_excel_vs_sql.py. Re-run with python scripts/validate_excel_vs_sql.py in the py313 conda env to refresh both the stdout output and the PDF.